

Automation

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Where are we now?

2.SDLC+ WhyDevOps

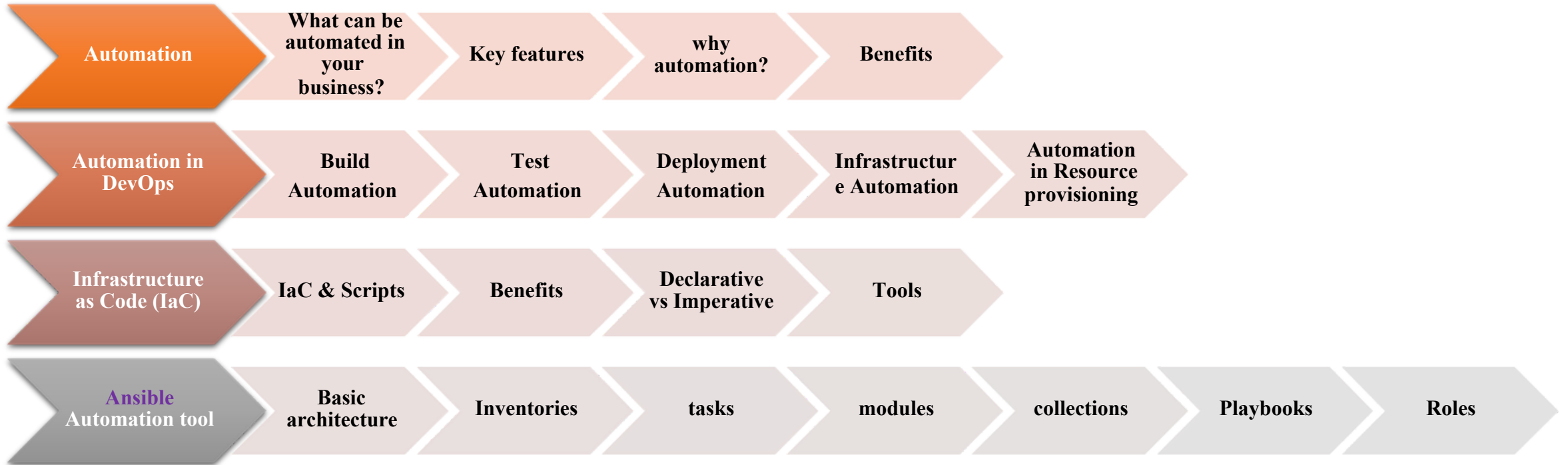
3.DevOps Phases + Cloud Computing –Basics

4.VCS –GitLab

6. Microservices

7. Automation – Ansible

OUTLINE



Automation

...performs tasks with reduced human assistance

What can be automated in your business?

- IT automation
- business automation
 - Business process management
 - Business process automation
- robotic process automation
- industrial automation
- AI/ML/Deep Learning
 - Adaptive algorithms that use predictive models to perform tasks without explicit instructions
 - E.g. *predictive maintenances*
 - Automatically detect repetitive manual tasks and later automate them
- Etc....

Key features of automation technologies

- Unified automation platforms
 - promote **consistency** and let you **standardize automated processes** and content across
 - development, test, and production environments.
- Easy, straightforward operations
 - allow more team members to participate and contribute.
- Overall simplicity
 - promotes greater adoption of automation throughout your organization.
 - E.g. Winery GUI, AWS data pipeline,
- Integration with other tools and products
 - allow you to more easily automate a broader range of tasks and components.
 - making it **highly pluggable**
- Platform scalability
 - allows you to simply expand the capacity of your pipeline as adoption grows.
 - E.g. scale according to the requests coming from mobile devices and laptop devices

...but why automation?

- Blue Cross and Blue Shield of North Carolina saved*
 - > US\$850,000
 - 70,000 work hours.
 - by automating VM provisioning
 - executed 200,000 Ansible Playbooks
- The City of Denver, without Ansible Automation Platform
 - 20 minutes to create an individual Microsoft Teams environment
 - *Less than a min* with automation platform
- Creating 'Autonomous' AI
 - src: <https://www.vice.com/en/article/qjkwqb/meta-ai-chief-publishes-paper-on-creating-autonomous-artificial-intelligence>

* <https://www.redhat.com/en/resources/bcbsnc-case-study>

...but why automation? -Benefits

- **Consistency**

- Consistent infra for both dev and ops teams

- **Scalability**

- Automatic scaling up/down

- **Speed**

- **Flexibility**

- **Standardization**

- Both dev and ops team will follow the same standard toolstack

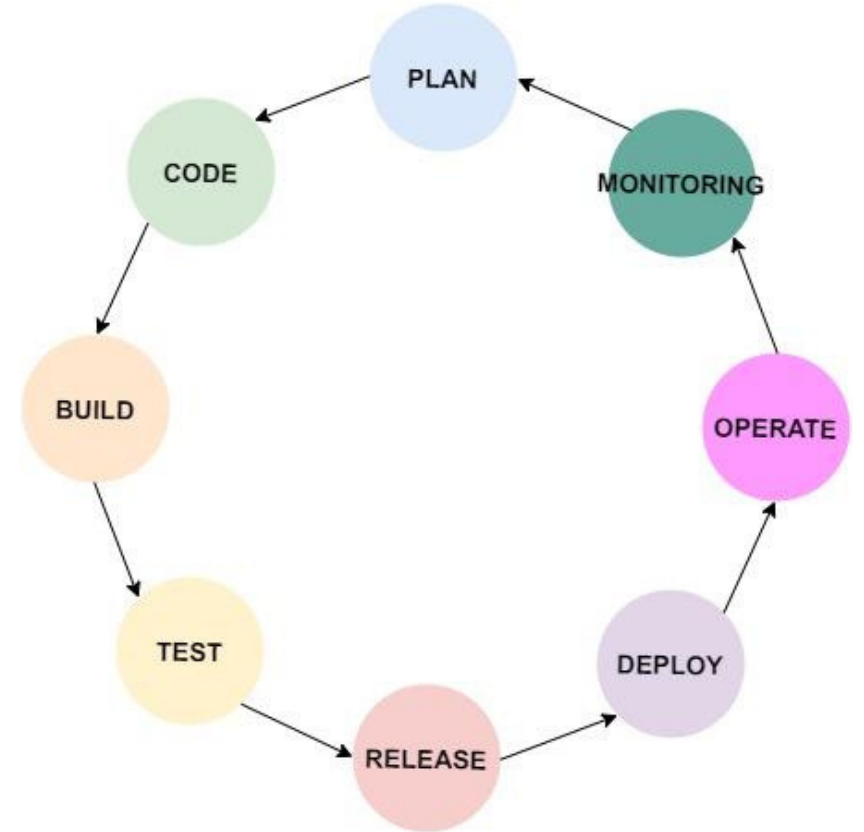
* <https://www.redhat.com/en/resources/bcbsnc-case-study>

Automation in DevOps

...use of technology to perform tasks with reduced human assistance [[src](#)].

helps you

- accelerate processes
- scale environments
- build continuous integration
- Continuous Testing
- build continuous delivery
- build continuous deployment (CI/CD) workflows

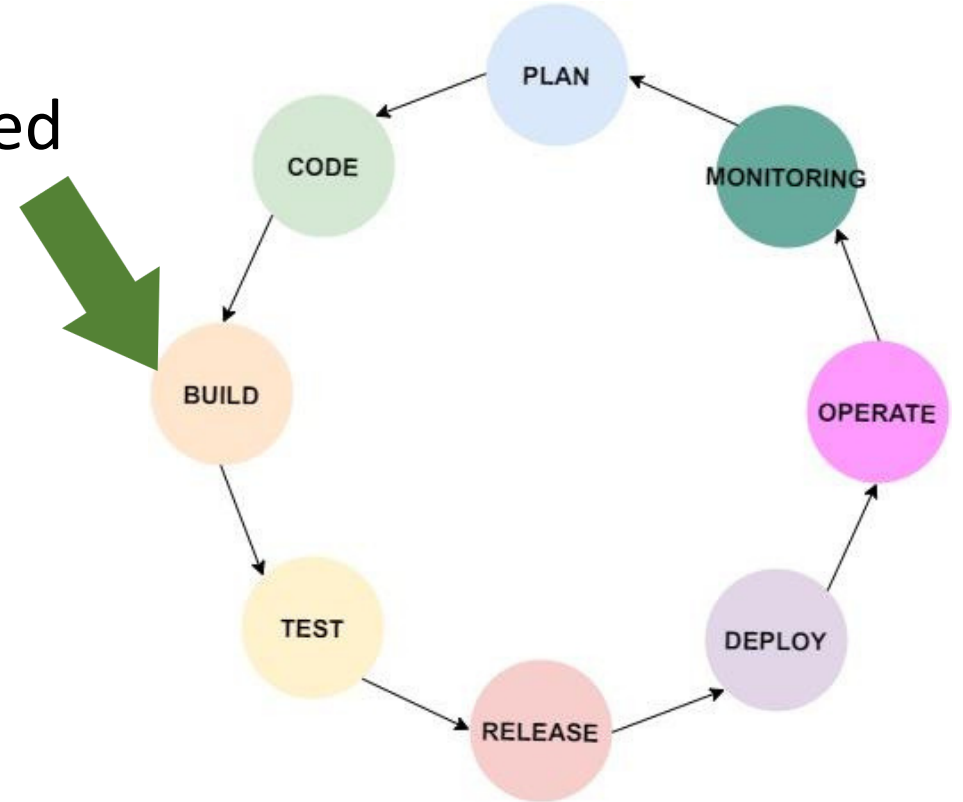


Build Automation in DevOps

- Automatic source code retrieval
- Source code compilation into binary
- Packaging compiled files (maybe into a compressed one)

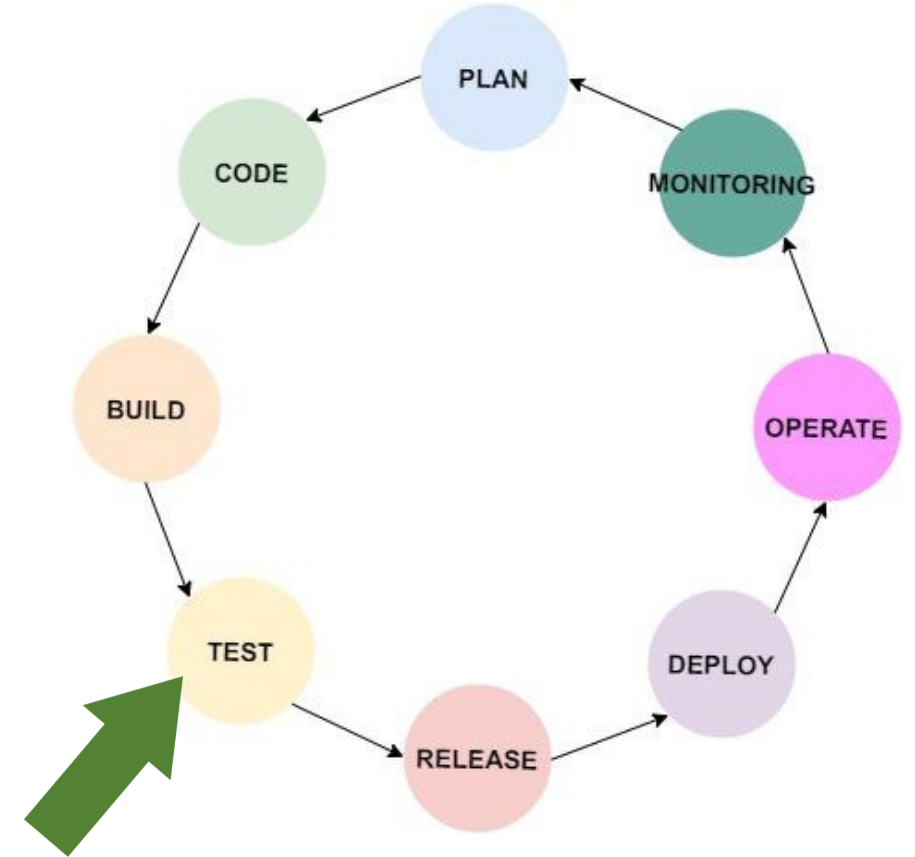
Key metrics

- Average build time
- Frequency of builds
- Success rate
 - x% success/failed



Test Automation in DevOps

- Unittests -> codeanalytics
- Integration tests
- System tests
- Installation test
- User experience tests
- UI tests **Key metrics**
- Test execution time
- Test code coverage

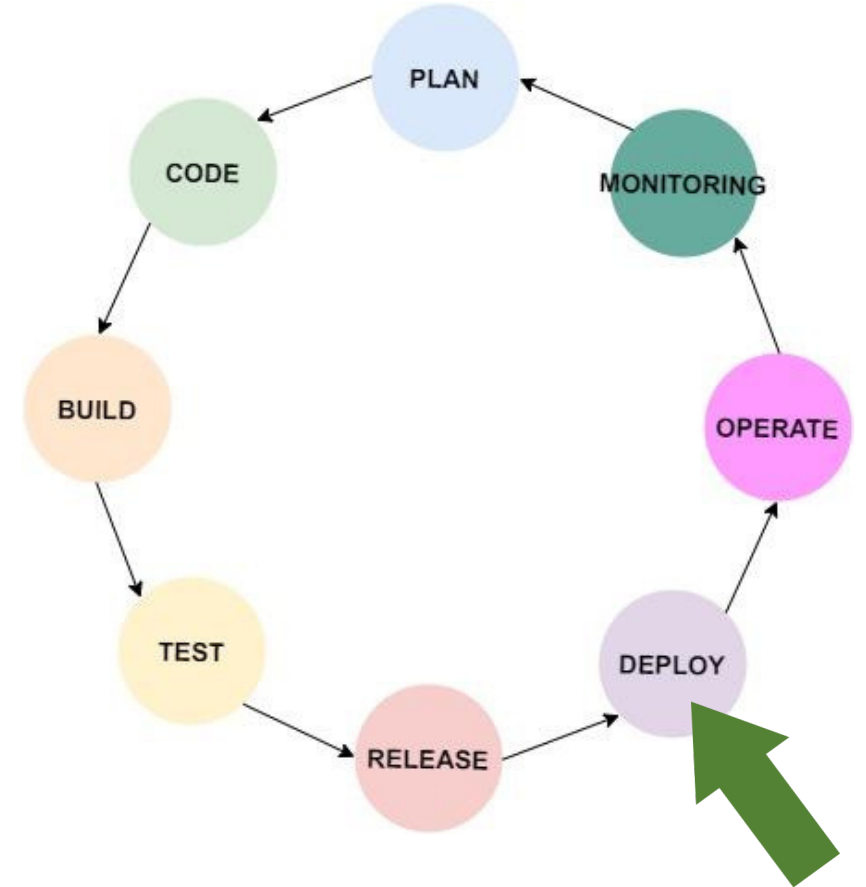


Deployment Automation in DevOps

- Model the environment
- Deploy the application

Key metrics

- % of failed/success deployments
- Avg. deployment time
- Deployment frequency
- Environment provisioning time

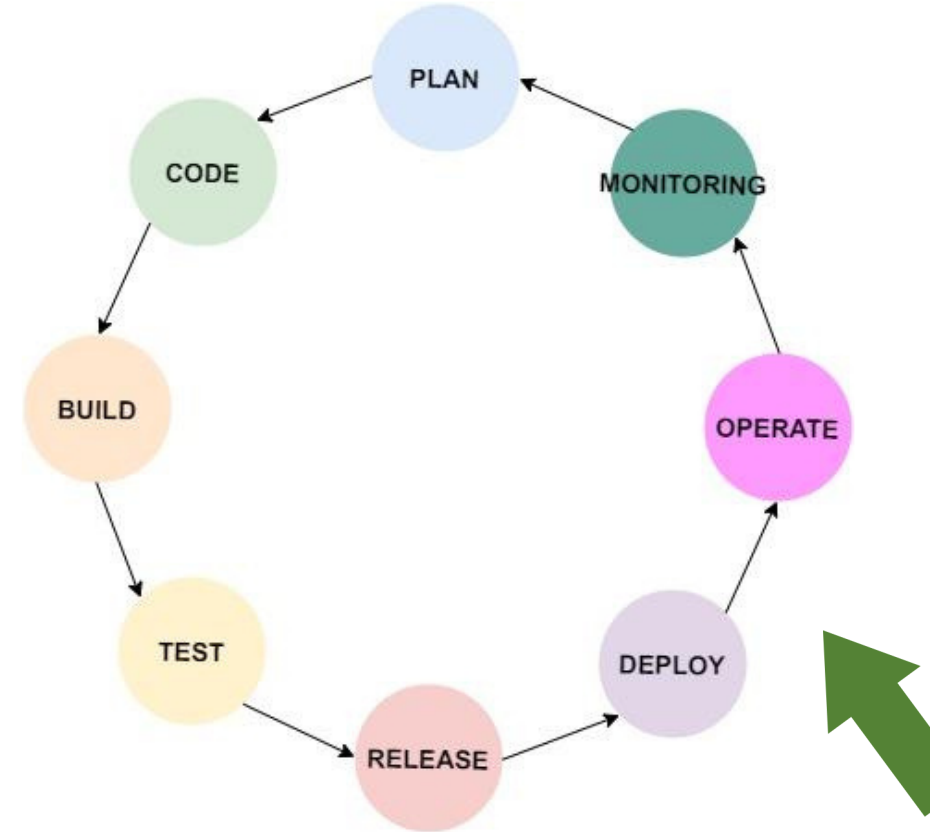


Infrastructure Automation in DevOps

- Create the required infrastructure
 - For development environment
 - For testing environment
 - For production environment

Key metrics

- Environment provisioning time
- Average deployment time
- Resource configuration
- Frequency of environment provisioning



Infrastructure as Code (IaC)

- Not a product
- A methodology
 - for rapid deployment of application
 - Preparing the environment
 - Configuration

Infrastructure as Code (IaC)

- No manual work to provision the infrastructure:
 - Create VM
 - Prepare specific OS image
 - Create specific network interface
 - Create storage server
 - Manually assigning computing resource to each VM
- E.g. setting up 1000 VMs for an application deployment may need more than 15 weeks and > 10 dedicated staffs
- IaC:
 - No more manual work
 - Write a code ->
 - Provision the required infrastructure and configure

Infrastructure as Code (IaC) –Examples

```
- name: Create VM
  os_server:
    state: present
    name: "{{ vm_name }}"
    image: "{{ image }}"
    key_name: "{{ key_name }}"
    flavor: "{{ flavor }}"
    network: "{{ network }}"
    security_groups: default
    boot_from_volume: true
    volume_size: 20
    timeout: 200
```

```
- name: Delete VM
  os_server:
    state: absent
    name: "{{ id }}"
```

```
- name: provision instance
  ec2:
    key_name: "{{ ssh_key_name }}"
    instance_type: "{{ instance_type }}"
    image: "{{ image }}"
    region: "{{ region }}"
    wait: true
    vpc_subnet_id: "{{ vpc_subnet_id }}"
    assign_public_ip: true
```

```
- name: terminate instance
  ec2:
    state: absent
    region: "{{ region }}"
    instance_id: "{{ id }}"
```


laC& Scripts

- Scripts
 - Not suitable for large scale system
 - Hard to keep track of
 - Hard to maintain
 - Used to automate a static step
 - Not flexible
- laC
 - It is also a script-like environment -> high-level scripting
 - Provides versatility of code
 - More human readable
 - Platform-agnostic

Infrastructure as Code (IaC) –Benefits

- **Cost** reduction Increase in **speed** of deployments
- Reduce **errors**-> risk
- Infrastructure/configuration
- **consistency**
 - Development and Operation teams are aligned
- Reduced **management** over head

Infrastructure as Code (IaC) –Tools

- Ansible
- Puppet
- Chef
- Terraform
- SaltStack
- AWS CloudFormation
- Azure Resource Manager
- Google Cloud Deployment Manager

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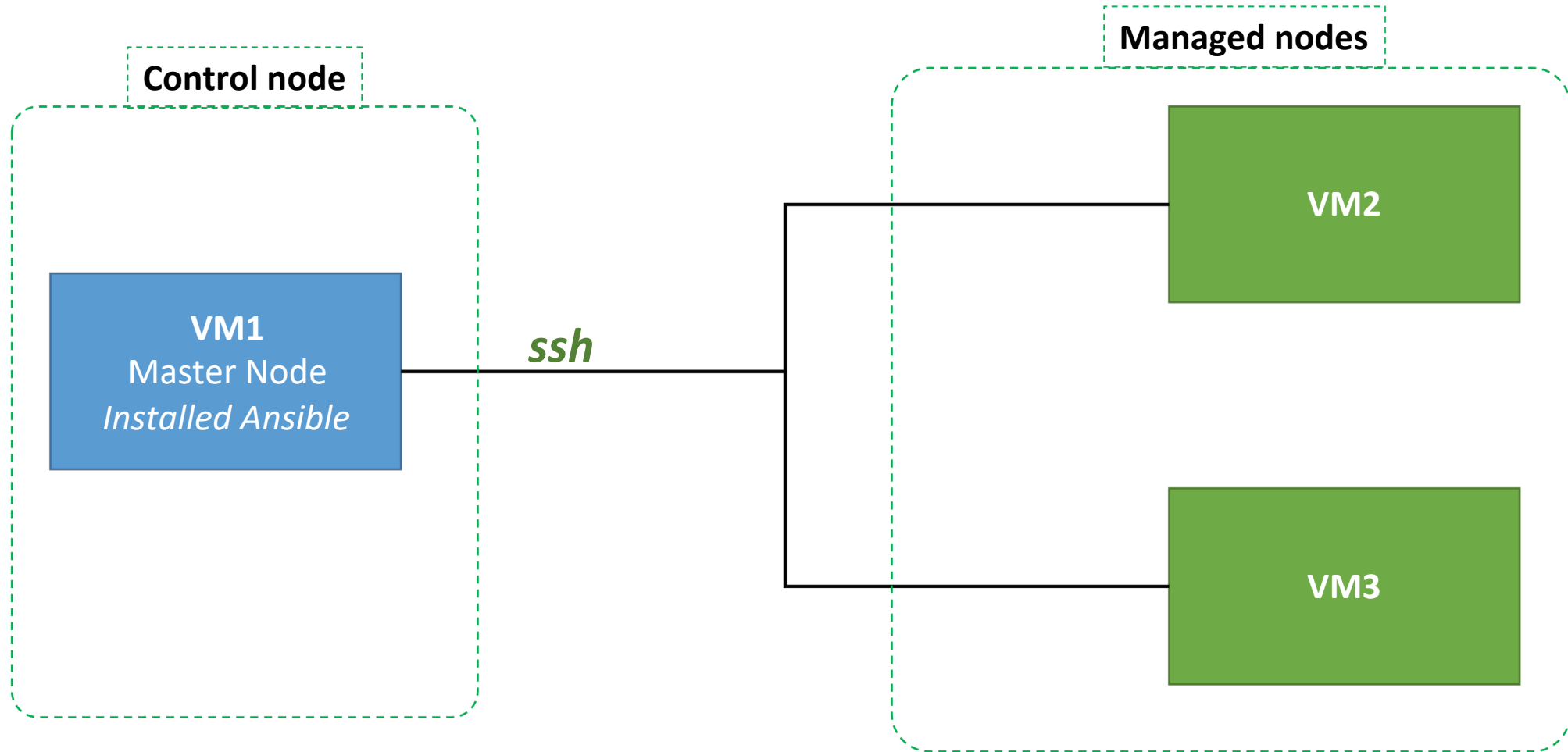
Ansible Automation tool

- Founded in 2013
- Bought by Red Hat in 2015.
- An **open source** *infrastructure automation* tool
- A **configuration** management tool
- Gives the ability to control and configure multiple servers from one single location
 - Avoid doing repeated tasks manually
 - E.g. Setting up the database server, webserver, etc
- **Agentless**
 - Require only ssh access
- Over 2500 modules (https://docs.ansible.com/ansible/2.8/modules/list_of_all_modules.html)
- Easy to learn
 - Because of so high level descriptive languages (using YAML)

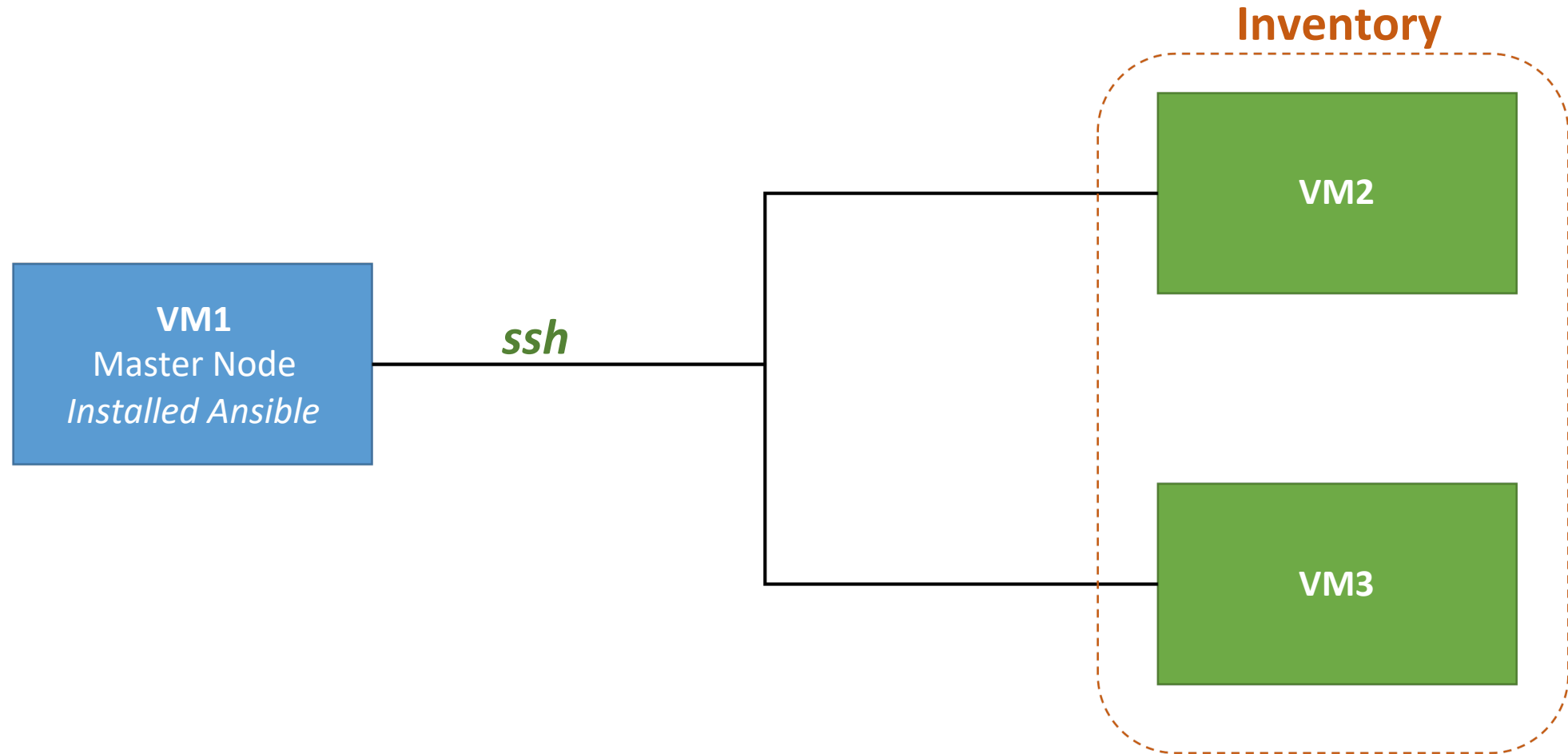
Why Ansible?

- Agentless
 - Target system requires only sshd and python
 - No daemons or agents to install
- Security
 - Relies on ssh
- Easy to get started, compared to the others!
 - Human readable, YAML format

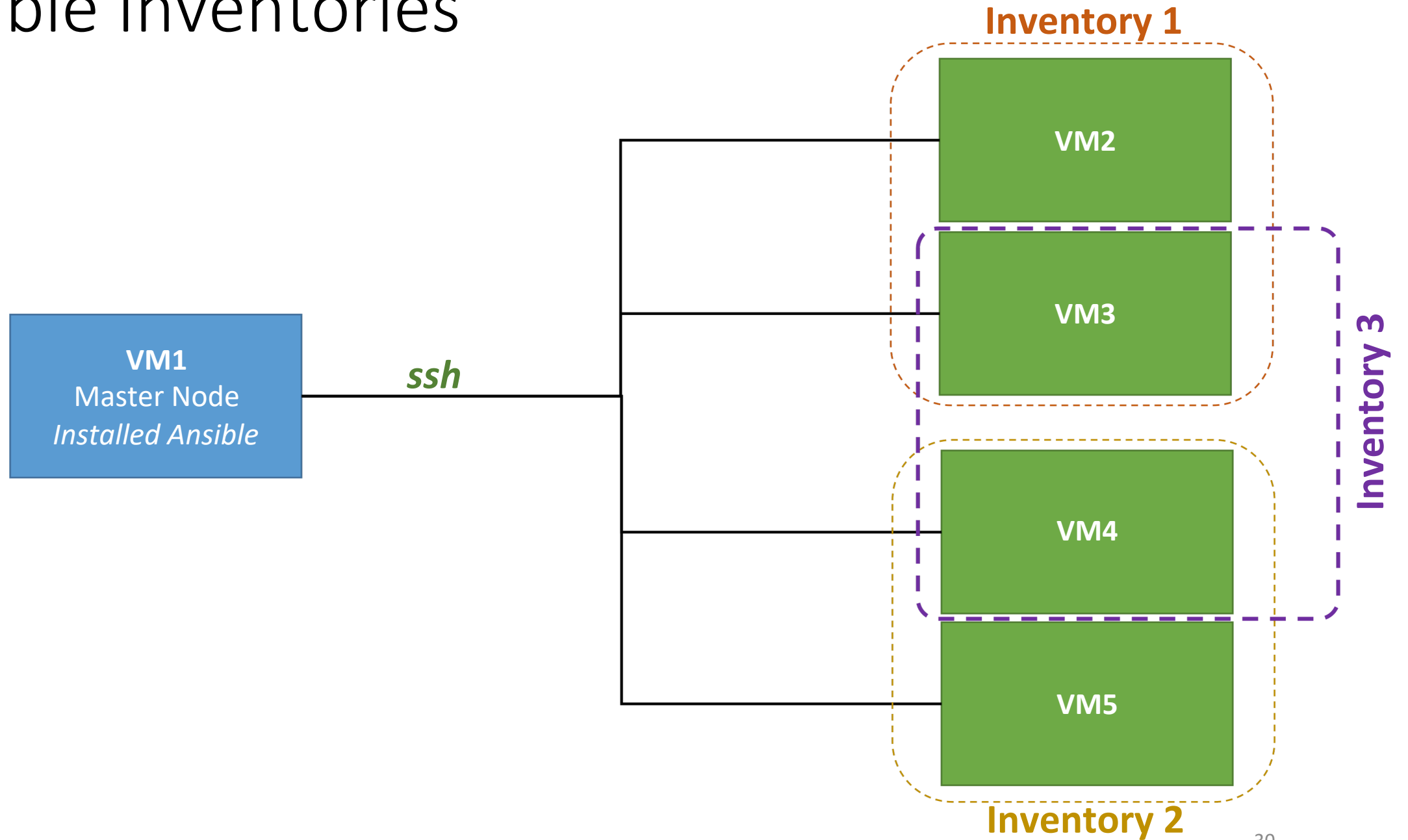
Basic Ansible Architecture



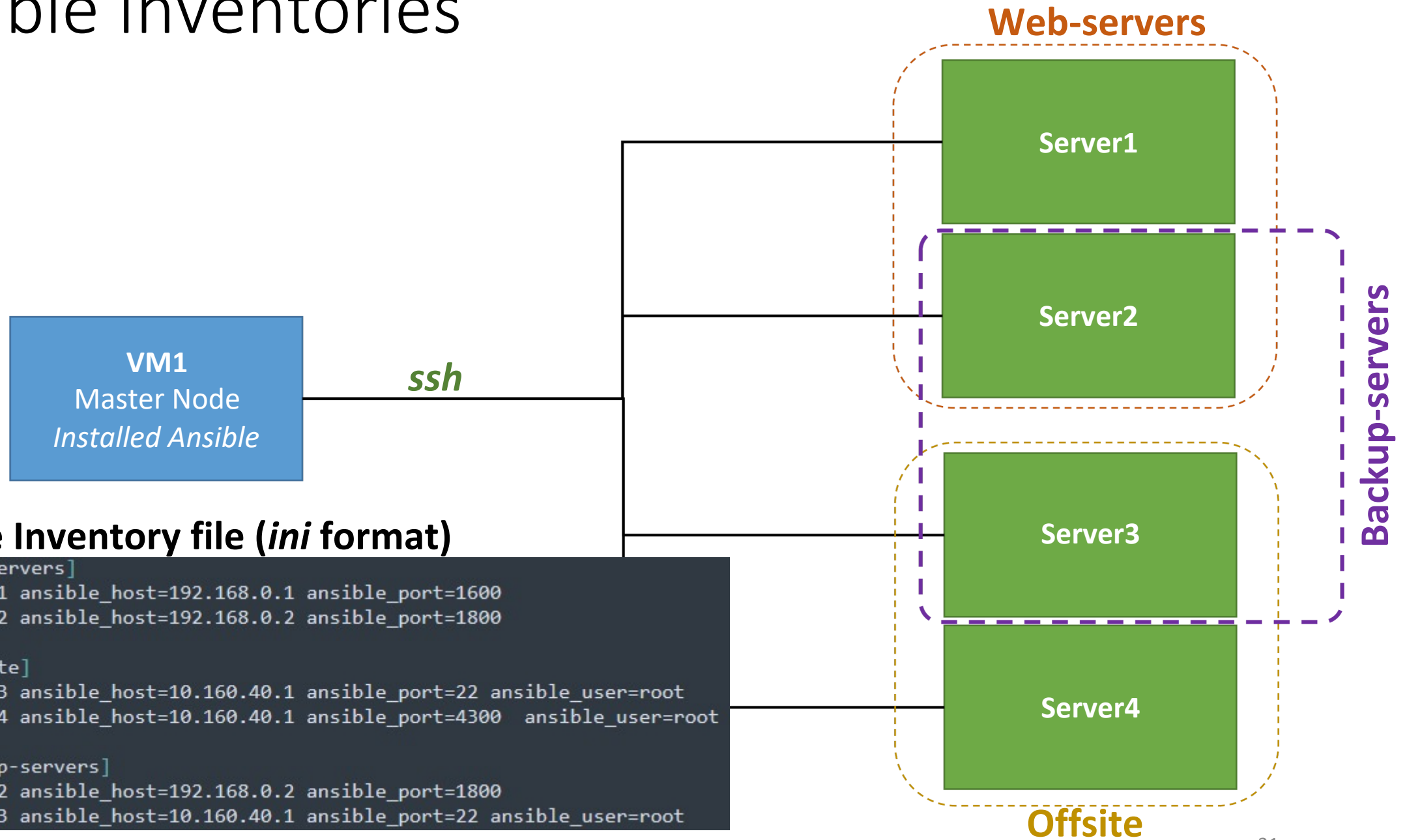
Ansible Inventories



Ansible Inventories



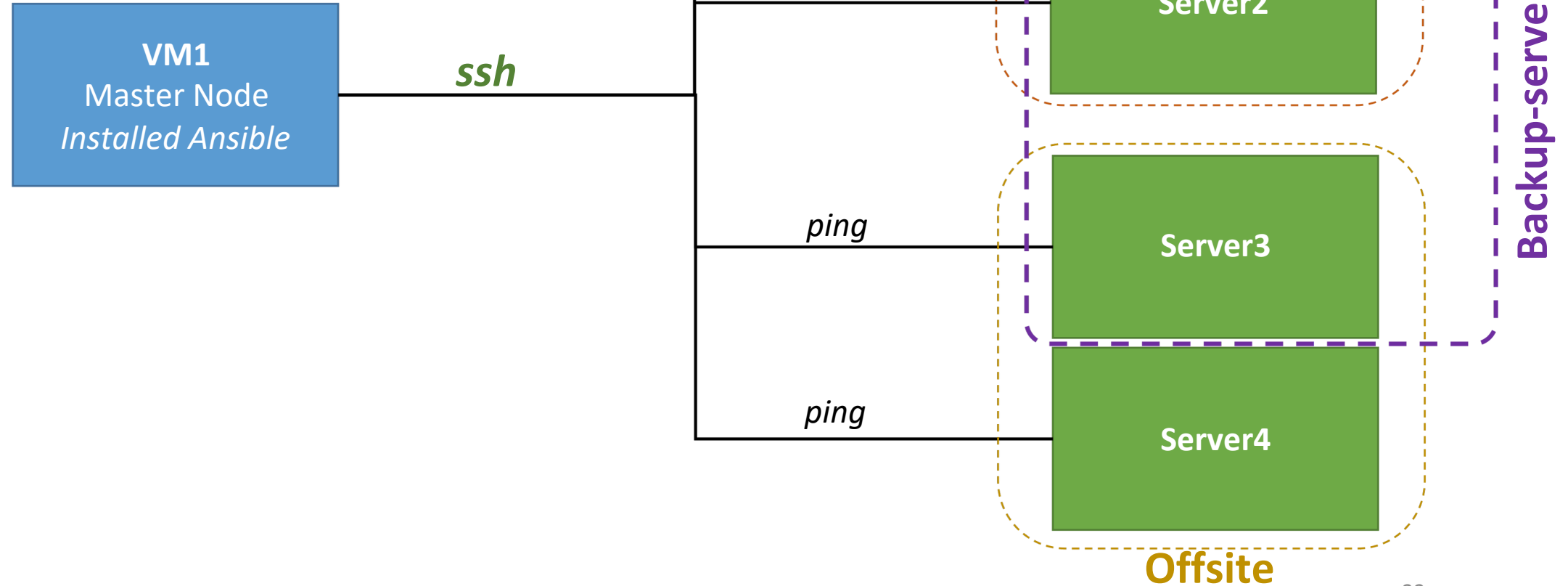
Ansible Inventories



Ansible tasks

Now, You want to test if all the servers are accessible from **VM1**

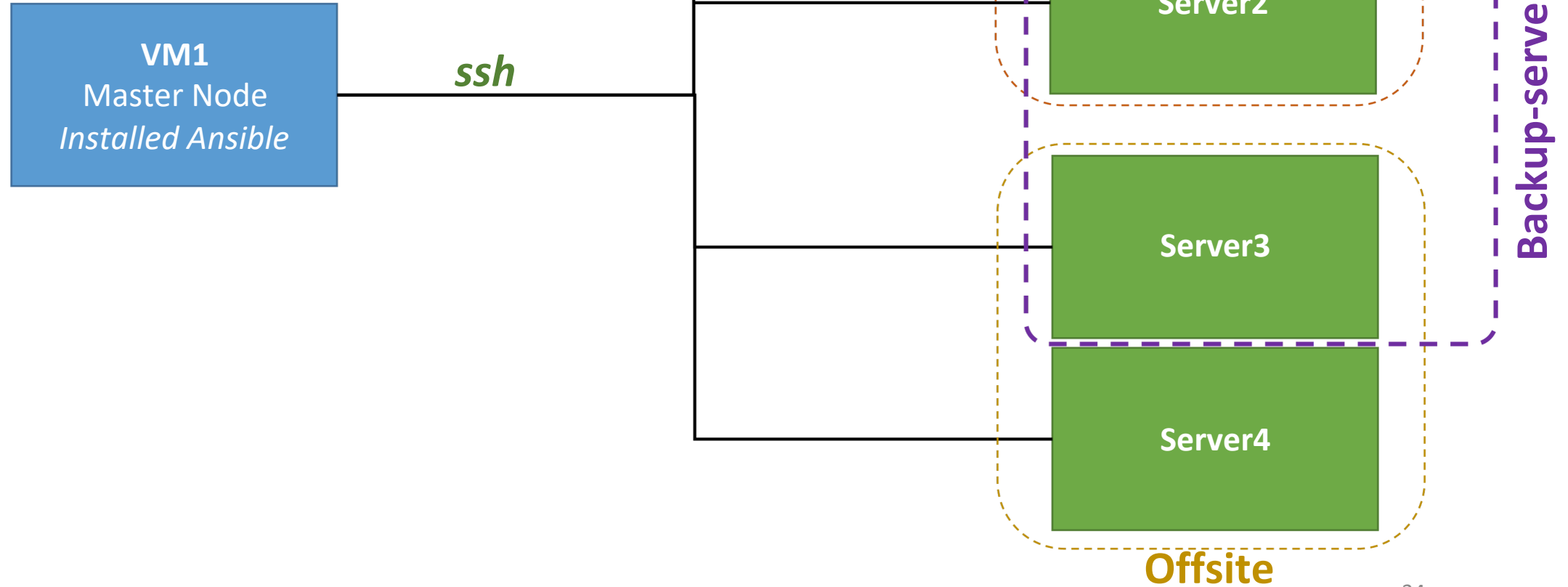
```
1 $ ansible all -m ping
```



Ansible tasks

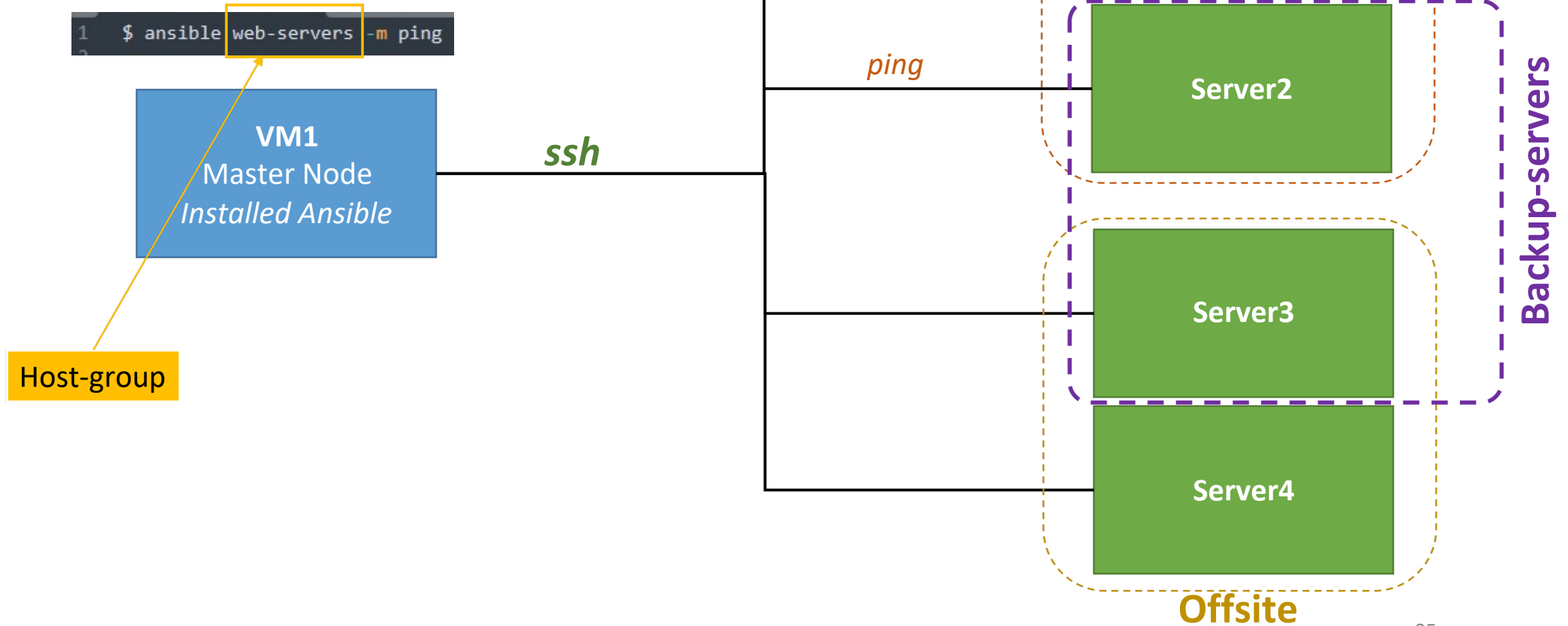
Now, You want to test if all the servers are accessible from **VM1**

```
1 $ ansible web-servers -m ping
```



Ansible tasks

Now, You want to test if all the servers are accessible from **VM1**



Ansible modules

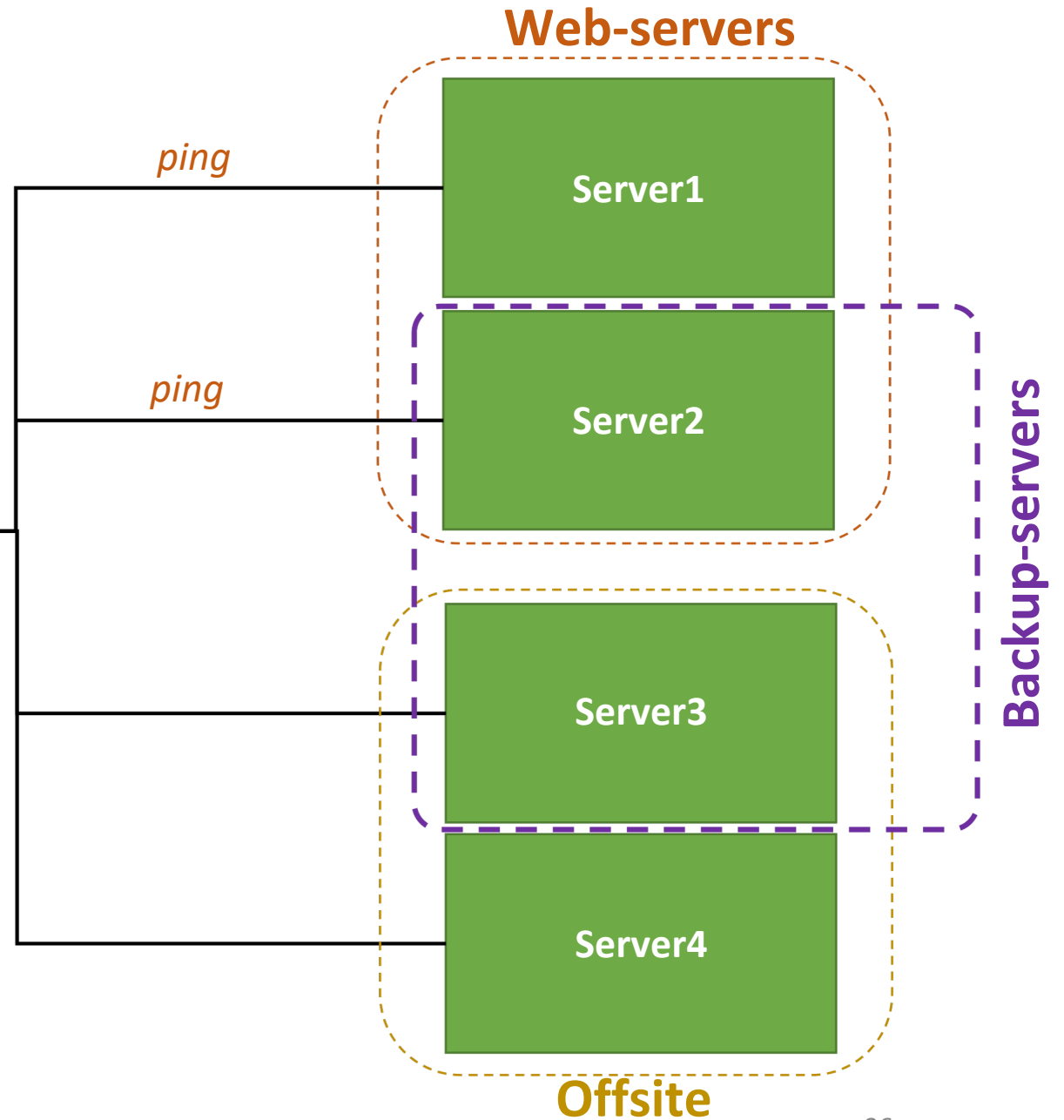
Now, You want to test if all the servers are accessible from **VM1**

```
1 $ ansible web-servers -m ping
```

VM1
Master Node
Installed Ansible

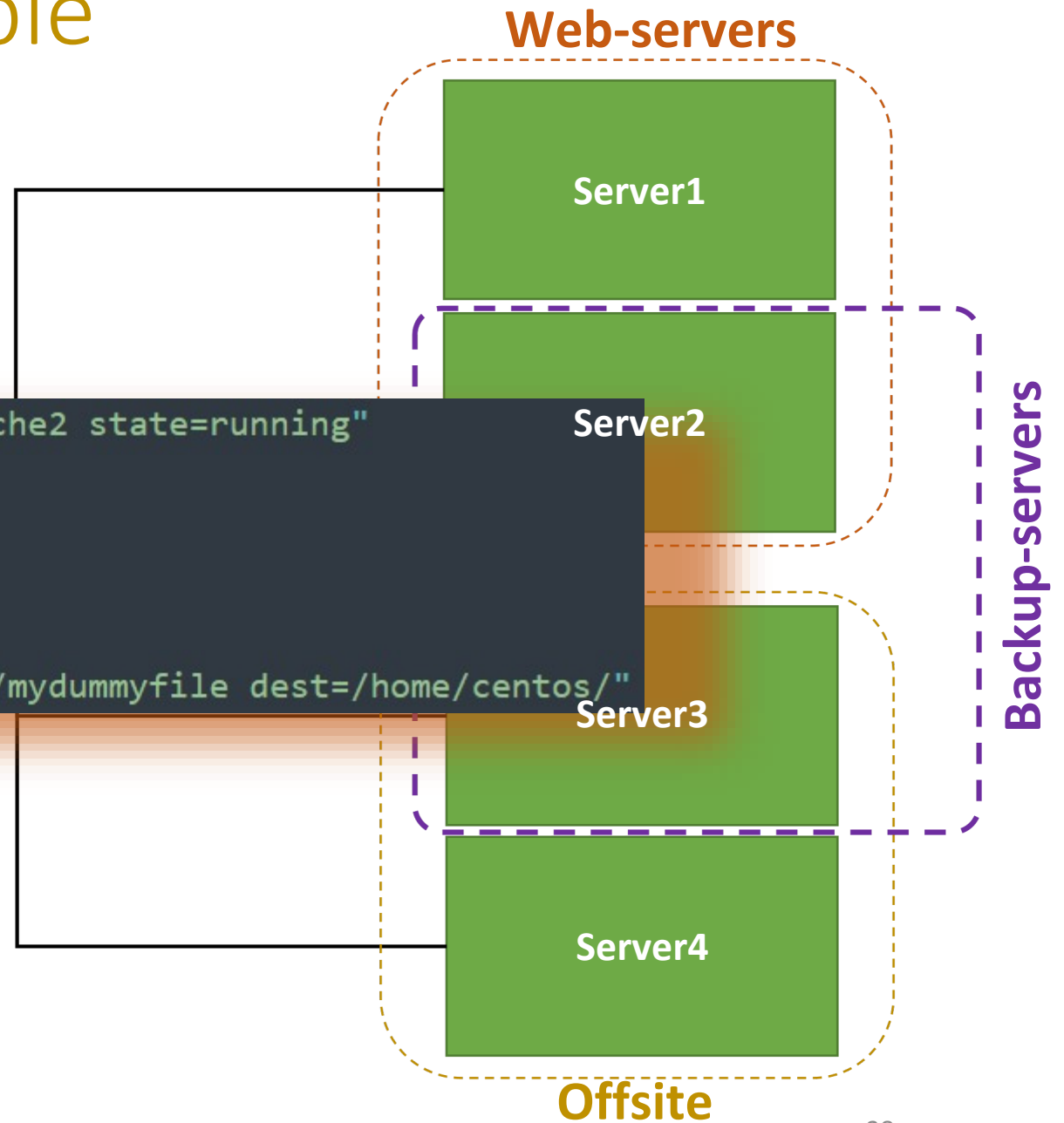
ssh

Module name



Ansible modules -Example

```
1 $ ansible web-servers -m service -a "name=apache2 state=running"
2
3 $ ansible offsite -m command -a "pwd"
4
5 $ ansible offsite -m command -a "ls"
6
7 $ ansible backup-servers -m copy -a "src=/tmp/mydummyfile dest=/home/centos/"
```



Ansible modules

- The units of code Ansible executes.
- Each module has a particular use.
- modules are grouped in **collections**

Ansible modules

- The units of code Ansible executes.
- Each module has a particular use.
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The List of collections

amazon.aws	ibm.qradar	community.hashi_vault
ansible.builtin	infinidat.infinibox	community.hrobot
ansible.netcommon	inspur.sm	community.kubernetes
ansible.posix	junipernetworks.junos	community.kubevirt
ansible.utils	kubernetes.core	community.libvirt
ansible.windows	mellanox.onyx	community.mongodb
arista.eos	netapp.aws	community.mysql
awx.awx	netapp.azure	community.network
azure.azcollection	netapp.cloudmanager	community.okd
check_point.mgmt	netapp.elementsw	community.postgresql
chocolatey.chocolatey	netapp.ontap	community.proxysql
cisco.aci	netapp.um_info	community.rabbitmq
cisco.asa	netapp_eseries.santricity	community.routeros
cisco.intersight	netbox.netbox	community.skydive
cisco.ios	ngine_io.cloudstack	community.sops
cisco.iosxr	ngine_io.exoscale	community.vmware
cisco.meraki	ngine_io.vultr	community.windows
cisco.mso	openstack.cloud	community.zabbix
cisco.nso	openvswitch.openvswitch	containers.podman
cisco.nxos	ovirt.ovirt	cyberark.conjur
cisco.ucs	purestorage.flasharray	cyberark.pas
cloudscale_ch.cloud	purestorage.flashblade	dellemc.enterprise_sonic
community.aws	sensu.sensu_go	dellemc.openmanage
community.azure	servicenow.servicenow	dellemc.os10
community.crypto	splunk.es	dellemc.os6
community.digitalocean	t_systems_mms.icinga_director	dellemc.os9
community.docker	theforeman.foreman	f5networks.f5_modules
community.fortios	vyos.vyos	fortinet.fortimanager
community.general	wti.remote	fortinet.fortios
community.google		frr.frr
community.grafana		gluster.gluster
		google.cloud
		hetzner.hcloud
		hpe.nimble

Ansible collections

Ansible.Builtin collections-> **ansible.builtin.file** is a module

Some functionalities of *file* module

- Change file ownership, group and permissions
- Give insecure permissions to an existing file
- Create a symbolic link
- Touch a file
- Create a directory if it does not exist
- Remove file (delete file)
- Recursively remove directory

```
- name: Change file ownership, group and permissions
  ansible.builtin.file:
    path: /etc/foo.conf
    owner: foo
    group: foo
    mode: '0644'
```

```
- name: Create a directory if it does not exist
  ansible.builtin.file:
    path: /etc/some_directory
    state: directory
    mode: '0755'
```

```
- name: Remove file (delete file)
  ansible.builtin.file:
    path: /etc/foo.txt
    state: absent
```

Ansible Playbooks

- Ordered lists of **tasks**,
 - saved so you can run those tasks in that order repeatedly.
- Written in YAML and are easy to read, write, share and understand.

Ansible Playbooks –an Example

First play

```
1  ---
2  - name: Update web servers
3    hosts: webservers
4    remote_user: root
5
6    tasks:
7      - name: Ensure apache is at the latest version
8        ansible.builtin.yum:
9          name: httpd
10         state: latest
11      - name: Write the apache config file
12        ansible.builtin.template:
13          src: /srv/httpd.j2
14          dest: /etc/httpd.conf
15
16 - name: Update db servers
17   hosts: databases
18   remote_user: root
19
20   tasks:
21     - name: Ensure postgresql is at the latest version
22       ansible.builtin.yum:
23         name: postgresql
24         state: latest
25     - name: Ensure that postgresql is started
26       ansible.builtin.service:
27         name: postgresql
28         state: started
```

Second play

Playbook1.yaml

Execute the playbook:

```
1  ansible-playbook Playbook1.yaml
```

Ansible Roles

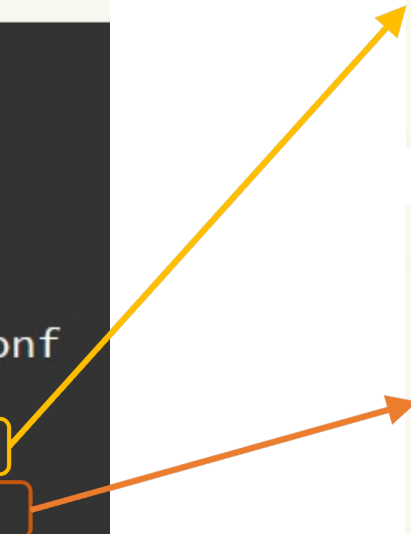
- Away of organizing tasks and related files to be later called in a playbook.
- Roles are a collection of tasks (not to confuse with Ansible collections),
 - moved from one playbook to another,
 - can be run independently but only through a playbook file.
- Roles can be considered as something similar to libraries

Ansible Roles –*An example*

- Away of organizing tasks and related files to be later called in a playbook.
- Roles are a collection of tasks,
 - moved from one playbook to another, can be run
 - independently but only through a playbook file.
- Roles can be considered as something similar to libraries

playbooks/03.yml

```
---
- hosts: webserver
  become: true
  become_method: sudo
  vars:
    file_path: /etc/qux.conf
  roles:
    - create-file
    - install-package
```



playbooks/roles/create-file/tasks/main.yml

```
---
- name: Create an empty file
  file: path={{ file_path }} state=touch mode=0644
```

playbooks/roles/install-package/tasks/main.yml

```
- name: Install a package
  apt:
    pkg: "{{ package }}"
    state: installed
    update-cache: yes
```

Idempotency in IaC

- Idempotent behavior :
 - Repeating the task does not change the final state
- Compare the “**current state**” with “**desired state**” of the system
- E.g. below “Create VM” task will not create multiple VMs if you run the task more than once
- Most Ansible modules are “Idempotent”

```
- name: Create VM
  os_server:
    state: present
    name: "{{ vm_name }}"
    image: "{{ image }}"
    key_name: "{{ key_name }}"
    flavor: "{{ flavor }}"
    network: "{{ network }}"
    security_groups: default
    boot_from_volume: true
    volume_size: 20
    timeout: 200
```


References

- <https://nsrc.org/workshops/2013/nsrc-ubuntunet-trainers/raw-attachment/wiki/Agenda/intro-ansible.pdf>
- <https://www.redhat.com/en>
- https://docs.ansible.com/ansible/latest/user_guide/playbooks_strategies.html
#playbooks-strategies
- [https://docs.ansible.com/ansible/latest/user_guide/playbooks_intro.html#abo](https://docs.ansible.com/ansible/latest/user_guide/playbooks_intro.html#about-playbooks)
ut-playbooks
- [https://searchitoperations.techtarget.com/feature/Ansible-vs-Chef-vs-Puppet-](https://searchitoperations.techtarget.com/feature/Ansible-vs-Chef-vs-Puppet-vs-SaltStack-A-comparison)
vs-SaltStack-A-comparison
- <https://www.redhat.com/en/topics/automation/what-is-devops-automation>
- <https://www.redhat.com/en/resources/bcbsnc-case-study>
<https://www.redhat.com/en/topics/automation>

THANK YOU